



ADVANCED RAG

73.69 - LARGE LANGUAGE MODELS - 2026

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REPASO

- ¿Cuál es el objetivo principal de RAG?
- ¿Cuáles son sus principales componentes? ¿En qué consiste el indexing pipeline?
- ¿Qué bases de datos vectoriales vimos?

01

ADVANCED RAG

Tipos de RAG



NAIVE
RAG

NAIVE RAG

RETRIEVAL

Chunks irrelevantes

Falta información importante

Disminuyen precision y recall

AUGMENTATION

Output incoherente

Información redundante

Solo repite y no procesa

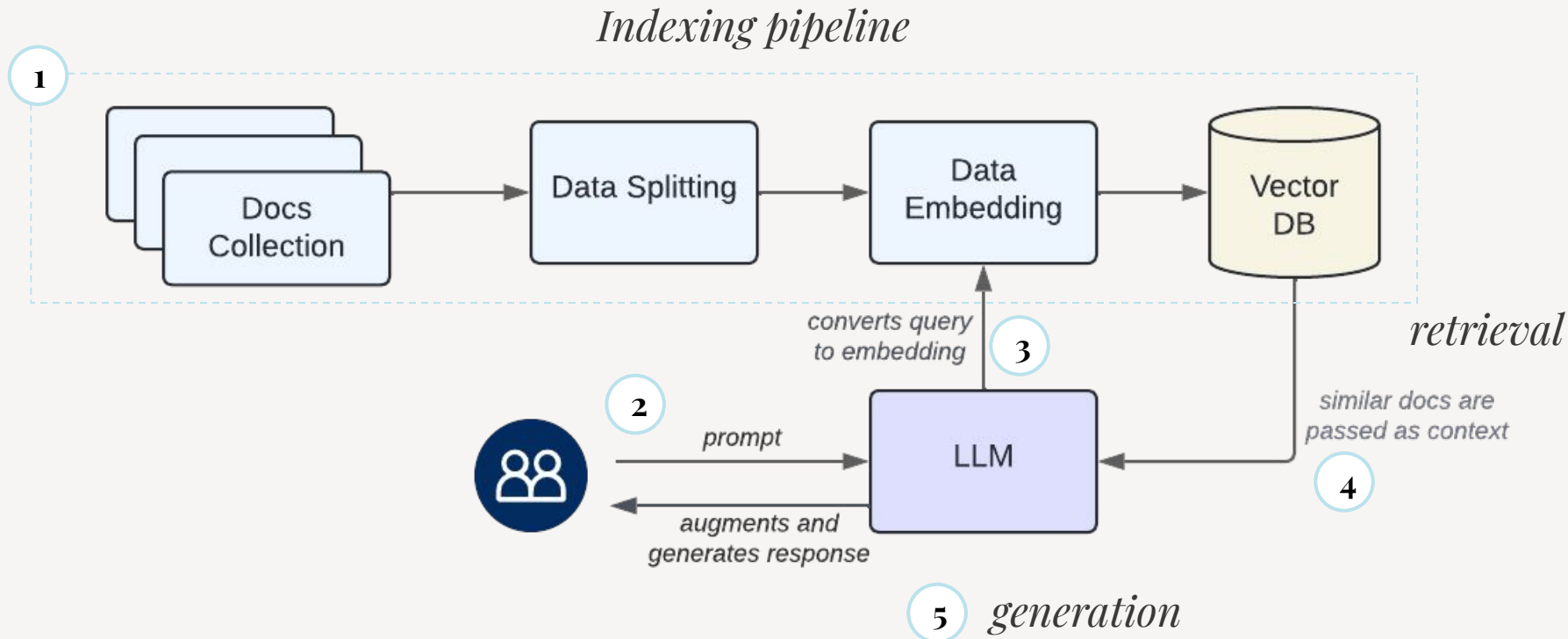
GENERATION

Aún puede alucinar si bien se mitiga

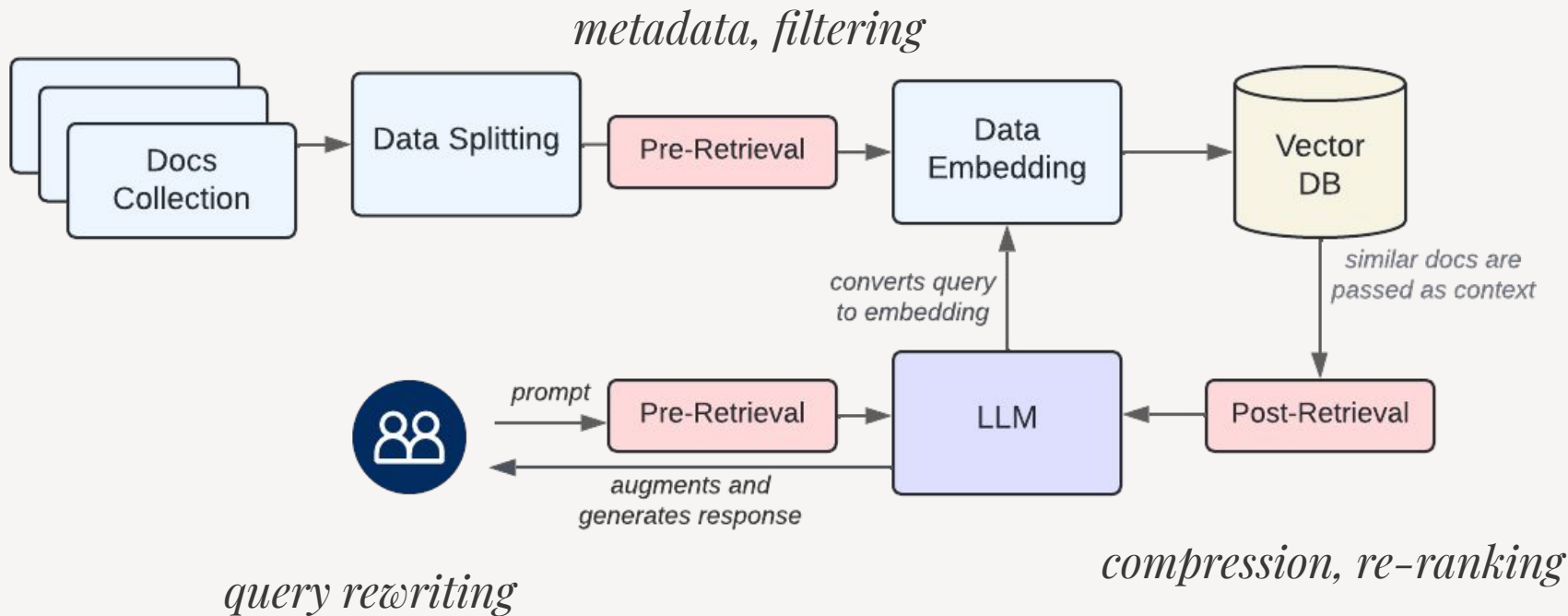
Información “biased”

Toxicity

PIPELINE NAIVE RAG



ADVANCED RAG



ADVANCED RAG - Metadata

PREPROCESSING

Agregar *metadata* a los chunks.

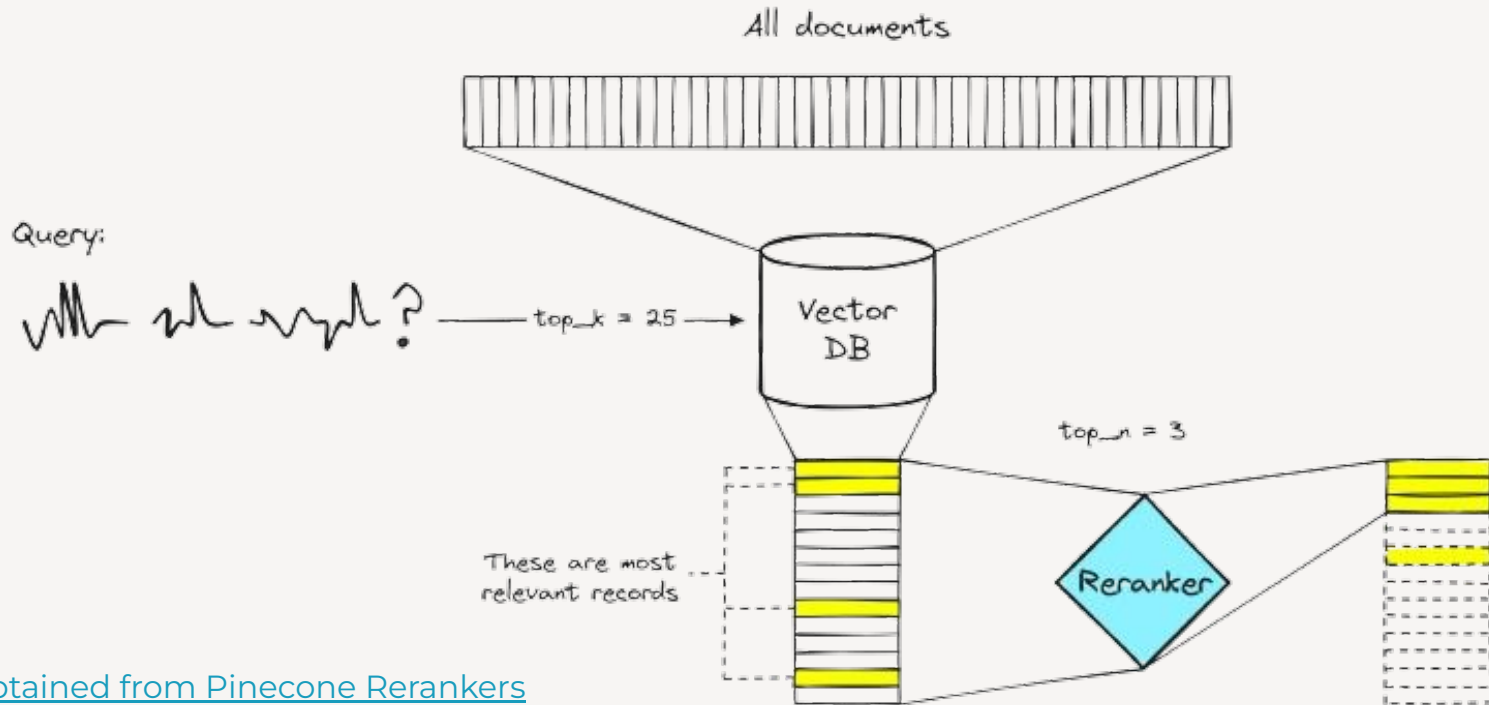
- Limita el scope del retriever permitiendo hacer filtros previos

Ejemplo: Número de página, autor, nombre del archivo, timestamp

```
{  
  "chunk": "Este es un fragmento de texto del documento.",  
  "metadata": {  
    "Número de página": 3,  
    "autor": "Juan Pérez",  
    "nombre del archivo": "documento_ejemplo.pdf",  
    "timestamp": "2024-08-14T12:34:56Z"  
  }  
}
```

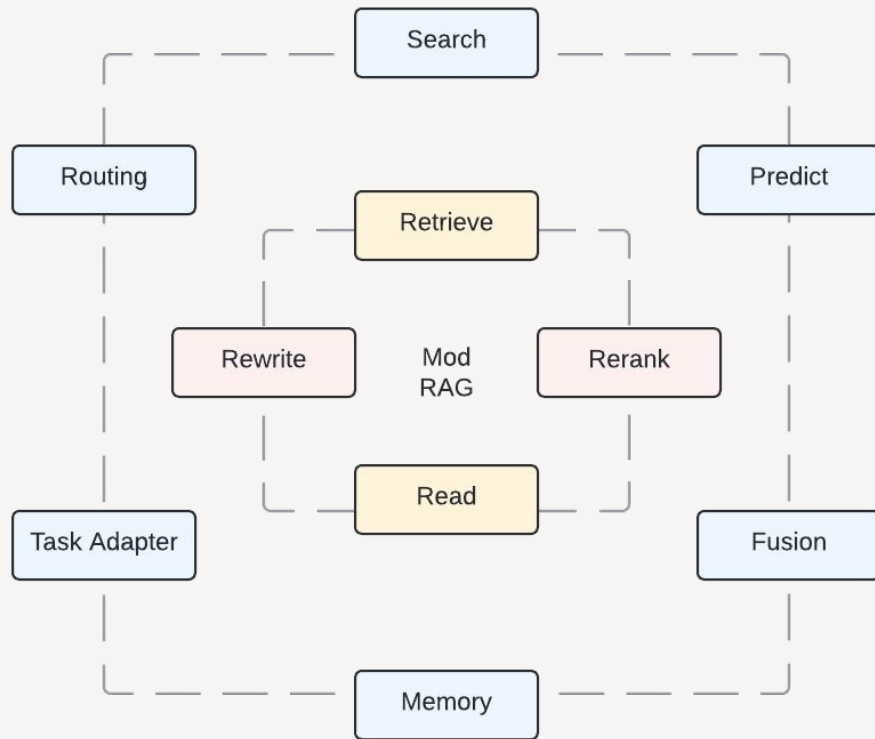
ADVANCED RAG - Reranking

“Context window” del LLM es limitado → priorizar

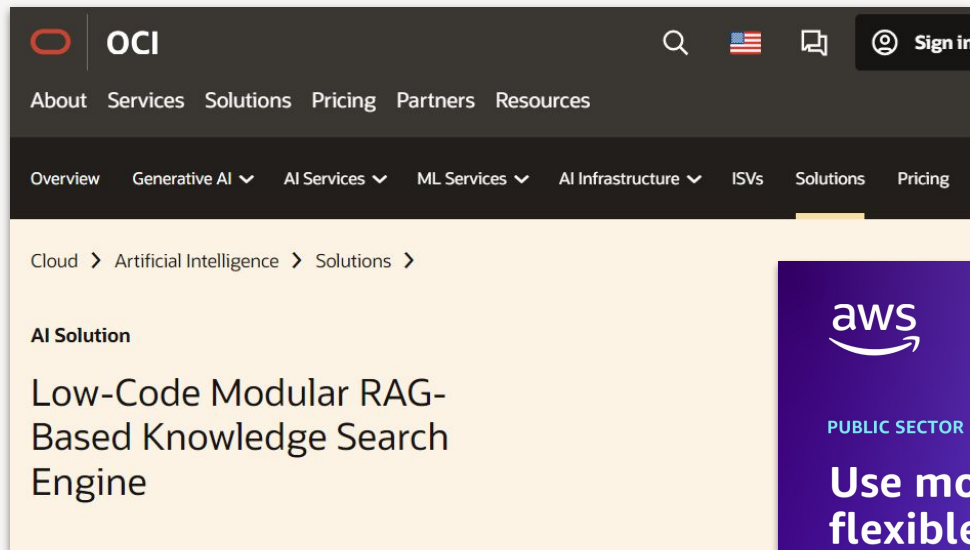


MODULAR RAG

- Agrega módulos con más técnicas de pre y post-procesamiento
- Permite flexibilidad y escalabilidad
- Facilita la adaptación y mejora de cada módulo de forma aislada.



MODULAR RAG



03

OTROS FRAMEWORKS Y APIs

Pytorch y Tensorflow

HUGGING-FACE

Existen distintos modelos de Transformers: GPT-2/3/4, BERT, RoBERTa, DistilBERT, T5

Transformers

Search documentation

Ctrl+K

V4.35.0

EN



114,893

GET STARTED

Transformers

Quick tour

Installation

TUTORIALS

Run inference with pipelines

Write portable code with AutoClass

Preprocess data

Fine-tune a pretrained model



Transformers

State-of-the-art Machine Learning for [PyTorch](#), [TensorFlow](#), and [JAX](#).



Transformers provides APIs and tools to easily download and train state-of-the-art pretrained models. Using pretrained models can reduce your compute costs, carbon footprint, and save you the time and resources required to train a model from scratch. These models support common tasks in different modalities, such as:



Natural Language Processing: text classification, named entity recognition, question answering, language modeling, summarization, translation, multiple choice, and text generation.



Computer Vision: image classification, object detection, and segmentation.



Audio: automatic speech recognition and audio classification.



Multimodal: table question answering, optical character recognition, information extraction from scanned documents, video classification, and visual question answering.

DEEP LEARNING FRAMEWORKS

Build, train y deploy de modelos de Deep Learning



MLOps CLOUD

PRODUCT	aws	Microsoft Azure	Google Cloud Platform
Machine learning	Amazon SageMaker	Azure Machine Learning	Google Cloud AI Platform
Image recognition	Amazon Rekognition	Azure Cognitive Services	Google Cloud Vision
Speech	Amazon Polly, Amazon Transcribe	Azure Cognitive Services	Google Cloud Speech-to-Text and Text-to-Speech
Natural language processing	Amazon Comprehend	Azure Cognitive Services	Google Cloud Natural Language API:
Big Data Analytics	Databricks	Databricks	Databricks
Chat Bot	AWS Chatbot	Azure Bot Service	Dialogflow

☰ Azure



Sign in

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Tools to build and scale generative AI applications

Innovate faster with new capabilities, a choice of industry-leading FMs, and infrastructure that pushes the envelope to deliver the highest performance while lowering costs.

[Explore more generative AI tools](#)

Service

Amazon Q

Customize this generative AI-powered assistant for the needs of your business

Service

Amazon Bedrock

Easily build and scale applications with LLMs, FMs, and generative AI tools

Service

Amazon SageMaker

Build, train, and deploy your own FM models at scale

Video y Paper sugeridos

Retrieval-Augmented Generation for Large Language Models: A Survey

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Abstract—Large Language Models (LLMs) showcase impressive capabilities but encounter challenges like hallucination, outdated knowledge, and non-transparent, untraceable reasoning processes. Retrieval-Augmented Generation (RAG) has emerged as a promising solution by incorporating knowledge from external databases. This enhances the accuracy and credibility of the

in Figure 1. The development trajectory of RAG in the era of large models exhibits several distinct stage characteristics. Initially, RAG's inception coincided with the rise of the Transformer architecture, focusing on enhancing language models by incorporating additional knowledge through Pre-Training Models (PTM). This early stage was characterized by work aimed at refining pre-training techniques. Subsequent arrival of ChatGPT [6] marked a turning point, with LLM demonstrating powerful in context capabilities. RAG research shifted towards providing information for LLMs to answer more complex, edge-intensive tasks during the inference stage, and development in RAG studies. As research on the enhancement of RAG was no longer limited to the pre-training stage but began to incorporate more with LLM inference techniques.

The emerging field of RAG has experienced swift growth,

Pregunta abierta

¿Qué atributos de seguridad se deberían considerar en RAG?

¿Qué vemos la clase próxima?



04

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BIBLIOGRAFÍA

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